

ITEC615 – INFORMATION TECHNOLOGY ESSENTIALS

Assessment 3: Cloud Migration Strategy for ABC Company

Semester 2, 2024

Submitted to:

Shakir Karim

NLIC, CL, NSY

Submitted by:

Amir Tamang

ID: S00393664

Email: S00393664@myacu.edu.au

Date of Submission: 09/11/2024

Feed Forward:

Reflection Questions	Feedback and Improvement Actions
1. In your opinion, how do you think previous project inputs contributed to the development of this report?	As stated in previous reviews, this needs a more thorough explanation of the technical decisions taken, which I have tried to cover in this report: for example, the considerations made against the costs, performance, and the meeting of objectives set by ABC Company in choosing Azure PaaS and parts of the hardware.
2. For future projects, what type of feedback would you find most helpful?	Detailed feedback is what helps me tune-especially those areas that need work, like technical detail, clarity, or structure. When I know exactly where I need to add evidence or revise my arguments, I can construct stronger, more targeted arguments in future assignments.
3. Were you having problems trying to understand the use of comments from previous assignments?	Some of the previous critique had been too general to locate specifically areas that needed attention. For instance, comments that highlight specific passages or ideas which require more careful consideration or evidence are of more value than the general advice to "improve the analysis."
4. How has your strategy changed considering earlier criticism?	Previous remarks emphasized that comments should be short and fact-based. Therefore, in this research work, I tried to give the shortest explanations and examples that will ensure that things are clear and support my recommendations effectively, such as comprehensive estimates of costs for cloud services.
5. About anything, how do you think the feedback process could be improved to help you learn more?	This would also be assisted if recommendations were added concerning readings or resources on hard subjects, such as cloud architecture models. Additionally, apart from these, feedback should include a summary of strengths and areas of development. That would really help me understand and lift the standard of my work in general.

Figure 1: Feed forward.

Checklist:

My submitted assignment report is within the specified word limit	✓
I have included references using specified referencing style	✓
I have correctly cited all my sources and references	✓
I have formatted my report as per the specifications	✓
I have checked my Turnitin report to ensure the similarity report is within the acceptable level(below 20% similarity)	✓
I have actioned feedback advice provided to me from previous assignment feedback	✓
I have completed proof reading and checked for spelling and grammar	✓
I have submitted my work before the due date/time	✓
I have submitted feed forward template along with my assignment submission	✓

Abstract

This paper presents a proposal to migrate to Azure Cloud to resolve operational issues regarding scalability, performance, and cost of increased volumes of traffic that ABC Company is currently experiencing with its e-commerce platform. The study will examine PaaS options within a hybrid cloud environment: one that balances the financial constraints of ABC with scalability and security needs.

The paper reviews cloud service paradigms-PaaS, IaaS, and SaaS-particular hardware and software components, and pricing considerations to develop an efficient migration plan. Emphasizing industry-specific requirements for e-commerce, data security, and customer happiness, the report compares the top cloud providers.

Microsoft Azure is the better fit based on the robustness of its PaaS offerings and compatibility with hybrid clouds. Its facilities of load balancing, auto scaling, and managed security of networks at Azure address performance issues at the company directly. The transition, although offering greater efficiency and lesser downtime, will require ABC Company to invest in specialist cloud training and probable changes for data-consuming activities.

These results show that a hybrid PaaS solution on Azure meets the performance, and scalability needs of ABC Company. Its resilience features, as well as auto-scaling, provide reliable e-commerce functions at reasonable costs in Azure.

Regarding the outcome, ABC Company is supposed to go ahead with Azure PaaS on a hybrid configuration basis. Certain technologies in Azure, such as VPN Gateway and Traffic Manager, should be employed wherever required for maximum network security and user experience. Employee training should be a priority so that the company itself can smoothly operate the implementation of cloud management processes.

The very focus of this report on Azure's offerings could have shortchanged the breadth needed to account for unique features offered by other cloud vendors. Additionally, cost estimates are rough estimates that may vary based on future actual usage and scalability needs.

Table of Contents

1.0. Introduction	1
1.1. Project background	1
2.0 Cloud Type and Platform.	2
2.1 Model for Migration	2
2.1.1 Platform as a Service (PaaS)	2
2.1.2 Hybrid Cloud Services.....	2
2.1.2 Cloud Provider and it is justification.	2
3.0 Hardware and Software Components.....	3
3.1 Selected Hardware components	3
3.2 Selected Software components	3
3.3 Justification of Selected hardware components.....	3
3.4 Justification of Selected Software Components.	4
4. Budget.....	4
5.0 Reliability and Expandability	5
6.0 Ethical, Security, and Sustainability Issue.....	6
7.0 Conclusion	6
References	7

1.0. Introduction

1.1. Project background

Unprecedented spikes of traffic during peak hours, between 9 am and 6 pm, give operational problems to the e-commerce platform of The ABC Company based in North Sydney. These performance bottlenecks, caused by a high volume of transactions peaking at 100 per second combined with 50,000 site hits per second, have resulted in long waiting times and reduced customer satisfaction. Because of the limited resources available for additional servers, moving to Azure Cloud is recommended as a solution. It would further leverage scalable and priced cloud services offered by Azure to enable dynamic resource allocation. The move is targeted at performance, customer experience, and operational efficiency (Gupta et al., 2021).

The aim of this report is to explore in detail the move of ABC Company to Microsoft Azure's PaaS model that would help improve user experience, reduce downtime, and increase operational efficiency. The report will cover the following areas:

- Recommended cloud migration model and type.
- Identify and justify the elements of the software and hardware.
- Deployment, operation, and maintenance budget estimates.
- Assessment of the system for expandability and reliability.
- Some of the common ones are sustainability, security, and ethical issues (Althathi, 2021).

The main objectives of this report are:

1. To establish a cloud migration model that will meet the scaling requirements for ABC Company.
2. To determine what are the proper resources-software and hardware.
3. Ensuring value for money in operational efficiency.

What this means is that, for ABC Company to be able to hold up its competition and handle customer satisfaction during peak hours, the need for migration to Azure cloud infrastructure is quite paramount. With a deployable scalable, adaptable, affordable solution, the business will resolve current performance issues and prepare itself for future growth.

2.0 Cloud Type and Platform.

2.1 Model for Migration

2.1.1 Platform as a Service (PaaS)

The authors advise on a model-PaaS on Microsoft Azure because the latter provides managed infrastructure with flexible scaling. Since Azure PaaS will manage operating systems, servers, and networking, ABC Company will then have the headroom to focus on application performance and customer experience (Wulf et al., 2021).

2.1.2 Hybrid Cloud Services

ABC will execute a hybrid approach, integrating on-premises computers with the Azure public cloud. In this scenario, scalability is performed in cloud resources while sensitive data is retained locally.

2.1.2 Cloud Provider and its justification.

Here, PaaS remains the suggested approach to migration within the Microsoft Azure ecosystem, considering limited resources in finance and scalability needs at ABC Company (Khatib Sulaiman Dalam No et al., 2024). In PaaS, the underlying infrastructure comprising servers, operating systems, and networking aspects is taken care of, which would leave the team free to focus on user experience and application management. Auto-scaling in PaaS by Azure dynamically allocates resources depending on the demand, which is integrally important for addressing network traffic during peak hours of operation. PaaS is cheap because it simplifies operations and allows quick deployment by outsourcing the maintenance of infrastructures.

The paper proposes a hybrid cloud model that offers flexibility and security. This approach will address the high-traffic transactions in the cloud while maintaining critical data locally through the scalability of the public cloud Azure with on-premises computers. In this way, the hybrid cloud architecture offers backup options for all vital applications aside from improving security and management over confidential information.

With affordable pricing, connectivity with Microsoft services, and a host of business-specific features like Azure Monitor for real-time analytics and Azure Traffic Manager for load balancing, the Azure Cloud is ideal. In addition, Azure provides reliability for e-commerce

mission-critical workloads, given its 99.9 percent uptime SLA for PaaS services (Gupta et al., 2021). ABC Company is very well set up for Azure PaaS in a hybrid architecture, addressing the present and future scaling.

3.0 Hardware and Software Components

3.1 Selected Hardware components

- **Virtual Machines:** D2s_v3 and D4s_v3 instances, with the ideal balance between CPU and memory, are recommended for regular and high-volume applications in Azure.
- **Storage:** The best storage option to achieve high throughput and low latency, both vital components for real-time transaction processing, is through Azure Managed Disc with Premium SSDs.
- **Load Balancer:** Azure Load Balancer will be used for dividing network traffic among virtual machines to realize peak performance.
- **Network Components:** Azure Firewall and Azure VPN Gateway are two prioritized services for safe network security and communication; this becomes highly essential in a hybrid approach.

3.2 Selected Software components

- **Operating System:** Windows Server 2022. Ubuntu Linux, in certain cases, is cheaper.
- **Database:** It will provide Azure SQL Database services that are managed and essential to ensure smooth integration of data; it also is relevant to large transaction volumes.
- **Application Software:** Global load balancing will be managed with the use of Azure Traffic Manager while deployment speed will be enhanced via Azure App Service.

3.3 Justification of Selected hardware components

- **Virtual Machines:** Azure strikes a perfect balance between the price to performance with its instance type, D2s_v3, which has 2 vCPUs and 8 GB of RAM, while the other one is D4s_v3, equipped with 4 vCPUs with 16 GB of RAM. These scalable virtual machines

during peak hours manage a range of workloads while promising smooth user experiences at low costs.

- **Storage:** Azure Managed Disc with Premium SSDs provides the high-throughput, low-latency storage required to process transactions in real time. This storage option will support the expansion of ABC Company through automatic scaling to meet demand.
- **Load Balancer:** The Azure Load Balancer distributes traffic among virtual machines to prevent overloads. It ensures peak performance.
- **Network Components:** VPN Gateway and Firewall: Azure Firewall prevents illegal access and ensures safety and legality. Azure VPN Gateway: Secure on-premises and cloud connectivity (Khan et al., 2021).

3.4 Justification of Selected Software Components.

- **Operating System:** Windows Server 2022 it is easier to support Microsoft tooling. Alternatively, Linux - glandularly Ubuntu - is the least expensive open-source option for web servers.
- **Database:** ABC supports an enormous volume of transactions, and with the scalability of Azure SQL Database in support of high availability, it ensures that management and backups of data are performed effectively.
- **Application Software:** It uses Azure Traffic Manager, which load balances globally to further reduce latency and increases the user experience. It uses Azure App Service because it is quite easy to deploy and scale.
- **Cost-effectiveness:** The components selected will be suitable for the ABC budget, providing scalable and cost-effective solutions to meet the performance requirements without over-specification.

4. Budget

The forecasted annual costs for ABC Company's cloud migration to Azure include the costs of deployment, operation, and management, reflected in the following table. Based on forecasted peak-hour traffic demands, the forecast includes all necessary services-virtual machines, storage, load balancing, and application services.

Item	Monthly Cost (AUD)	Annual Cost (AUD)
Virtual Machines (D2s_v3)	\$400	\$4800
Storage (Premium SSD)	\$300	\$3600
Azure SQL Database	\$500	\$6000
Load Balancer	\$100	\$1200
VPN Gateway	\$120	\$1440
Azure Firewall	\$160	\$1920
Azure Traffic Manager	\$80	\$960
Total	\$1660	\$19,920

Table 1: Budget of ABC company.

By considering this financial constraint, these estimates ensure a valid estimate. The budget ensures that the system would be scalable to manage high demand loads.

5.0 Reliability and Expandability

The Azure architecture shall be dependable; load balancing and automated scaling will manage peak-hour traffic. It ensures continuous availability through data redundancy by using geographically dispersed data centers. It is easy to scale with Azure to support additional resources like virtual machines, storage, and more, which may be needed in meeting future requirements.

The proposed setup on Azure uses load balancing and scaling to manage heavy traffic during peak hours, ensuring reliability. During unforeseen events of disruption, native redundancy with geographically dispersed data centers is enforced to continue the availability of Azure services. Similarly, because Azure's platform is highly scalable, ABC Company can add virtual machines, storage, and any other services whenever the need arises for the future. The disaster recovery and backup options will include Azure Site Recovery to provide even more resiliency and protection against data loss. This is a cost-performance-balanced solution that ensures reliability and scalability.

6.0 Ethical, Security, and Sustainability Issue

Addressing ABC Company's move to the cloud requires addressing key issues in sustainability, security, and ethics for a transition that will sustain user confidence and data integrity, while being environmentally responsible.

- **Ethics:** due care of the data of consumers. Adherence to strict legislation concerning data privacy, such as the GDPR, would protect information about the clients and prevent misuse. Clear-cut permissions by users and transparency of data policies would instill customer trust into ethical norms for standards.
- **Security - Cybersecurity** to protect the information of clients and corporate functions. Azure grants rich security features such as firewalls, IAM through Azure Active Directory, and encryption of data in flight and at rest. For better adherence to industry security requirements, multi-factor authentication with regular security audits provides even better security from cyber threats and unauthorized accesses.
- **Sustainability:** Moving to Azure will help ABC Company enhance sustainability because some of the world's Microsoft runs most energy-efficient data centers, some of which are powered from renewable energy. With fewer on-premises servers required, ABC Company can reduce carbon footprint. Resource optimization features that are built into Azure include automated scaling, meaning resources are allocated only when required, which helps in the reduction of overall energy consumption.

7.0 Conclusion

Thus, migrating ABC Company's e-commerce to Azure cloud provides a secure and economical way of scaling up to accommodate high traffic needs. ABC Company can securely manage large volumes of transactions while ensuring that compliance with data security is guaranteed using the PaaS capabilities of Azure during a hybrid setup. The reliability of the proposed architecture is guaranteed by load balancing, automated scaling, and redundancy of data. Sustainability is further supported by the greener infrastructure of Azure. Microsoft Azure would be the ideal choice for this transfer, as it manages to equilibrate cost and environmental responsibility with technological performance by supporting the expansion and operation requirements of ABC Company.

References

- Althati, C., 2021. Machine Learning Solutions for Data Migration to Cloud: Addressing Complexity, Security, and Performance, *Aus. J. ML Res. & App.*
- Gupta, B., Mittal, P., Mufti, T., 2021. A Review on Amazon Web Service (AWS), Microsoft Azure & Google Cloud Platform (GCP) Services. *European Alliance for Innovation n.o.*
<https://doi.org/10.4108/eai.27-2-2020.2303255>
- Khan, D., Jung, L.T., Hashmani, M.A., 2021. Systematic literature review of challenges in blockchain scalability. *Applied Sciences (Switzerland)*.
<https://doi.org/10.3390/app11209372>
- Khatib Sulaiman Dalam No, J., Ibrahim Khalaf Jajan, K., M Zeebaree, S.R., 2024. Optimizing Performance in Distributed Cloud Architectures: A Review of Optimization Techniques and Tools. *Indonesian Journal of Computer Science Attribution* 13, 2024–2327.
- Wulf, F., Dresden, T.U., Lindner, T., Westner, M., Regensburg, O., Strahringer, S., 2021. IaaS, PaaS, or SaaS? The Why of Cloud Computing Delivery Model Selection-Vignettes on the Post-Adoption of Cloud Computing.